Title – Concentrations of Solutions

Extended Writing -

1. 48.0 g of methanol (CH_3OH) reacts with 72.0 g of oxygen to produce 66.0 g of carbon dioxide and 54.0 g of water.
2. 1.62 g of hydrogen bromide (HBr) reacts with 0.98 g of sulfuric acid (H_2SO_4) to produce 0.36 g of water, 0.64 g of sulfur dioxide (SO_2) and 1.60 g of bromine (Br_2).

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DO NOW

Independent practice review

1. $2CH_3OH + 3O_2 \rightarrow 2CO_2 + 4H_2O$
2. $2HBr + H_2SO_4 \rightarrow SO_2 + Br_2 + 2H_2O$

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DO NOW

Concentrations of Solutions

Lesson 8

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Complete the GC Quiz

Due 16th Nov

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HOME LEARNING

Learning objectives:

By the end of the lesson you will be able to:

- relate mass, volume and concentration
- calculate the mass of solute in solution
- relate concentration in mol/dm^3 to mass and volume.

Key Words

- **solute**
- **solvent**
- **solution**
- **concentration**

What is concentration?

What does the word concentration mean?

Why is the drink on the right more concentrate

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What is concentration?

Concentration is a measure of how many particles of solute are dissolved in a given volume of solvent.

The given volume is usually 1 dm^3 (1000cm^3).

A **more concentrated** solution has **more particles** per dm^3.

The unit for concentration is normally g/dm^3.

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Conversions

- 1 dm^3 = 1000 cm^3

  to convert from cm^3 to dm^3 you need to divide the value by 1000

- 1 cm^3 = 1/1000 dm^3 = 0.001 dm^3
- What is 20 cm^3 in dm^3?

  20 / 1000 = 0.02 dm^3

- What is 38 cm^3 in dm^3?

  38 / 1000 = 0.038 dm^3

Conversions

$1000\ cm^3 = 1\ dm^3$

to convert from dm^3 to cm^3 you need to multiply the value by 1000

What is 0.5 dm^3 in cm^3?

$0.5 \times 1000 = 500\ dm^3$

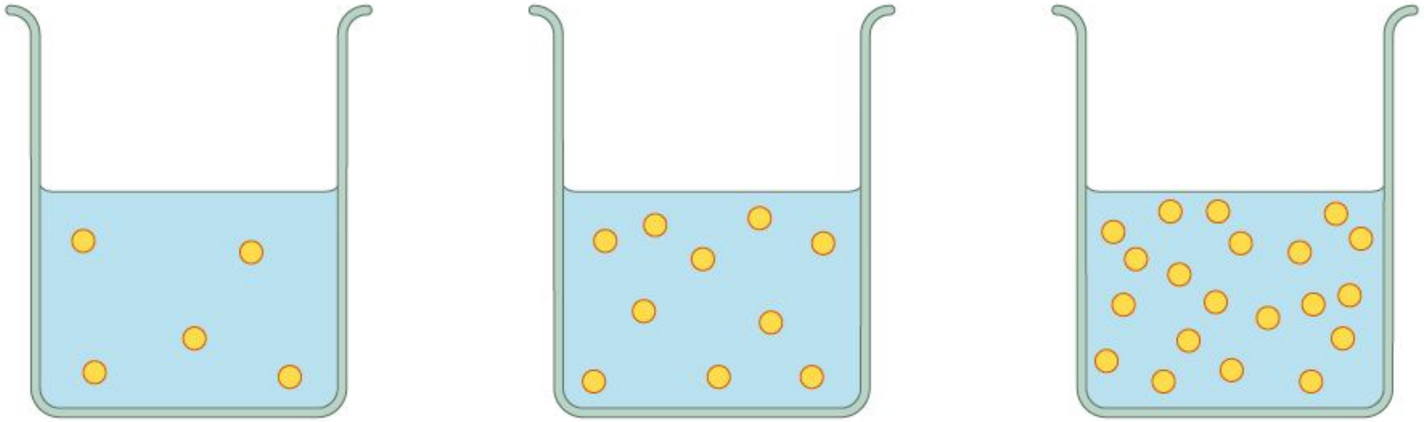
What is 3.8 dm^3 in cm^3?

$3.8 \times 1000 = 3800 cm^3$

Calculating concentration

$$\text{Concentration } g/dm^3 = \frac{\text{mass of solute (g)}}{\text{volume of solution } (dm^3)}$$

Remember to convert to dm^3 if you are given values in cm^3



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Calculating concentration example

Calculate the concentration of a solution if 13 g is dissolved in 250 cm^3 water.

- Convert 250 cm^3 to dm^3
- 250 / 1000 = 0.25 dm^3

Concentration = 13 / 0.25 = 52 g/dm^3

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Calculating concentration example

Calculate the concentration of a solution if 13 g is dissolved in 250 cm^3 water.

13 g in 250 cm^3

X 4

X 4

Step 1 -
convert to 1 dm^3
(1000 cm^3)

1000 cm^3

Step 2 -
do the same
to the mass

52 g

Concentration = 52 g/dm^3

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Another example

Calculate the concentration of a solution containing 2.1 g in 25 cm^3.

- Convert 25 cm^3 to dm^3
- 25 / 1000 = 0.025 dm^3

Concentration = 2.1 / 0.025 = 84 g/dm^3

Another example

Calculate the concentration of a solution containing 200 mg in every 10 cm^3.

Convert 200 mg to grams

200 / 1000 = 0.2 g

Convert 10 cm^3 to dm^3

10 / 1000 = 0.01 dm^3

Concentration = 0.2 / 0.01 = 20 g/dm^3

Independent practice

Calculate the concentration of the following in g/dm^3:

1. 40 g solute in 800 cm^3
2. 0.08 g solute in 20 cm^3
3. 90 g solute in 780 cm^3
4. 2.5 g of solute dissolved in a 500 cm^3 solution
5. 2.3 g of solute in a 250 cm^3 solution
6. 10 mg of solute in a 25 cm^3 solution
7. 15 mg of solute in a 750 cm^3 solution

Independent practice review

Calculate the concentration of:

1. 40 g solute in 800 cm^3 50 g/dm^3
2. 0.08 g solute in 20 cm^3 4 g/dm^3
3. 90 g solute in 780 cm^3 115 g/dm^3
4. 2.5 g of solute dissolved in a 500 cm^3 solution 5 g/dm^3
5. 2.3 g of solute in a 250 cm^3 solution 9.2 g/dm^3
6. 10 mg of solute in a 25 cm^3 solution 0.4 g/dm^3
7. 15 mg of solute in a 750 cm^3 solution 0.02 g/dm^3

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CONSOLIDATION